

Stavan Rupareliya

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Professional Summary

Embedded Software Engineer with 6+ years of experience in Edge AI, Embedded Linux, and SoC platforms. Transitioning into Design Verification with a strong foundation in Digital Design and Verilog HDL. Skilled in Python, Shell scripting, low-level debugging, and hardware-software integration. Currently building expertise in Verilog, constrained-random verification, and functional coverage.

Tech-Stacks

Technology	VLSI Design, Verification, SoC/ASIC Verification, Semiconductor, Embedded Systems, Artificial Intelligence, Computer Vision, Edge Computing
Concepts	ASIC Design/Verification, CPU Design, Computer Architecture, Digital Logic Design, Functional Verification
Languages	Verilog, C, C++, Python, Shell
Protocols	I ² C, SPI, UART, CAN, USB
Tools	Git, GitHub, GDB, Valgrind, Docker
Operating Systems	Linux, Embedded Linux, Windows

Professional Experience

Sr. AI Research Engineer L1,
Mantra Smart Identity (Mantra Softech), Ahmedabad, India.

Dec 2025 – Present

- Designed, developed, and optimized computer vision models and GStreamer-based media pipelines for an Edge AI product, owning the end-to-end model lifecycle — development, training, conversion/optimization, and deployment — to enable real-time video analytics on embedded platforms.
- Conducted market research to analyze competitor products and industry trends, providing insights that guided product strategy and feature development.

Senior Software Engineer,
Tata Elxsi Limited, Bangalore, India.

Mar 2025 – Oct 2025
Client : Comcast Corporation

- Developed AI/ML-based text and image summarization pipelines for multimedia applications on embedded platforms for Comcast, leveraging deep learning models for efficient content understanding and compression.
- Deployed and optimized AI inference pipelines on edge devices within embedded Linux environments, with a focus on performance profiling, debugging, and resource optimization.

Software Engineer,
elnfochips (An Arrow Company), Ahmedabad, India.

Jul 2021 – Mar 2025

- Designed and implemented a video analytics solution using C/C++ and Python, integrating deep learning and computer vision models with embedded hardware for real-time edge inference. Led hardware-software co-design efforts, bridging embedded systems with AI pipelines for seamless integration and deployment.
- Applied quantization and pruning techniques to computer vision models, achieving high accuracy with reduced computational overhead. Built a camera simulator with a modular image processing pipeline, facilitating rapid prototyping and client POC validation.
- Collaborated directly with U.S. and European clients, collaborating across engineering, product, and client teams to deliver solutions within agile development cycles.

Project Trainee Engineer
elnfochips (An Arrow Company), Ahmedabad, India.

Jan 2021 – Jun 2021

- Completed training in C++, Python, embedded systems, and Linux internals with hands-on system programming and memory management projects.
- Contributed to client deliverables and assisted in building future project PoCs.

Technical Projects

FIFO Design & Verification | Standalone RTL Verification Project

SystemVerilog; Open-source EDA

- Designed a parameterized synchronous FIFO for arbitrary depths with explicit full/empty boundary contracts; verified using an independent scoreboard, assertions, and deterministic random stress.
- Automated a five-configuration sign-off matrix with Icarus simulation, Verilator lint, Yosys synthesis checks, GitHub Actions, VCD waveforms, and retained regression evidence.

Standalone repository: github.com/stavanrupareliya36/fifo-design-verification

UART Transmitter & Receiver | RTL Design

Verilog; EDA Playground

- Designed a UART transmitter and receiver in Verilog RTL with configurable baud rate, implementing FSM-based control for start/stop bit framing, serial-to-parallel data conversion, and parity generation/checking for reliable serial communication.
- The simulated design was validated on EDA Playground using directed testcases and loopback testing, verifying data integrity across multiple baud rates and error conditions such as framing and parity errors.

Binary Artificial Neural Network | ASIC Design

Verilog; ModelSim, Synopsys

- Designed an RTL logic single-layer Binary Convolutional Neural Network which constituted a max-pooling layer on a randomized input matrix and kernel matrix extracted from 2 SRAMs using Verilog.
- The simulated design was validated using ModelSim. Synthesis was performed by using Synopsys Design Compiler to achieve minimum area and maximum throughput.

Certifications

- **Digital Design – Hands-on** Maven Silicon
Hands-on training in digital logic: combinational & sequential circuits, FSMs, and memories.
- **Verilog HDL – Hands-on** Maven Silicon
Practical RTL design using Verilog: combinational & sequential logic, FSMs, and simulation.
- **VLSI System On Chip Design – Overview** Maven Silicon
Overview of SoC design flow: architecture, IP integration, RTL design, and verification.
- **Verification Methodology Overview** Maven Silicon
Fundamentals of verification: SystemVerilog, UVM concepts, testbench architecture, and coverage.

Invited Talks & Guest Lectures

Nirma University, Institute of Technology | Expert Speaker for B.Tech Students

Jan 2024

- Delivered an Industry Expert Lecture on **Embedded AI & Computer Vision Applications** at **Nirma University, Institute of Technology**, covering real-time AI deployment, embedded optimization, and edge intelligence.

Nirma University, Institute of Technology | Expert Speaker for M.Tech Students

Aug 2024

- Delivered an Industry Expert Lecture on **Edge Computing and Computer Vision** for **M.Tech students specializing in VLSI and Embedded Systems**.

eInfochips (An Arrow Company) | Nirma University STTP

Sep 2024

- Represented eInfochips (An Arrow Company) as an Expert Lecturer at the Faculty Development Program (FDP), Institute of Technology, Nirma University, Ahmedabad, delivering a session on **"Edge Computing and Computer Vision"** focused on NVIDIA Edge AI.

Indore Institute of Science and Technology (IIST), Indore | Adratim Adhiratham 2.0

Jun 2026

- Invited to deliver an expert talk on **"Edge AI"**, covering AI hardware accelerators, computer vision, deployment pipelines, model optimization, and real-world Edge AI applications.

Nirma University, Institute of Technology | STTP on Edge AI and Vision

Jul 2026

- Delivered an invited expert talk on **"Edge AI and Vision"** under the STTP program, presenting **"Deep Dive into Edge AI: Concepts to Deployment"** to researchers and faculty members.

Technical Publication

Journal Paper 2020	The visual computer, An International Journal of Computer Graphics, Springer Publication, 2020 "Real-time detection and identification of plant leaf diseases using convolutional neural networks on an embedded platform", Springer Publication. Paper Link
Conference Paper 2021	2nd International Conference on Communication, Computing and Industry 4.0 (C2I4) , 2021. "Real-time Face Mask Detection System on Edge using Deep Learning and Hardware Accelerators", IEEE Publication. Paper Link
Conference Paper 2021	Proceedings of the International e-Conference on Intelligent Systems and Signal Processing, 2021. "Real-Time Tomato Detection, Classification, and Counting System Using Deep Learning and Embedded Systems", Springer Publication. Paper Link
Conference Paper 2022	Advances in VLSI, Communication, and Signal Processing, 2022 "Face Mask Detection and Counting Using Deep Learning and Embedded Systems", Springer Publication. Paper Link

Awards & Recognitions

Core Value Award - Customer First 2024

- Recognized for exceptional dedication to prioritizing customer needs, resulting in increased customer satisfaction and loyalty. Demonstrated an outstanding ability to address customer concerns and ensure a positive customer experience.

Pat On The Back Award 2022

- Honored for exemplary performance and significant contributions to an image processing project. Demonstrated exceptional leadership, teamwork, and problem-solving skills, consistently exceeding expectations.

Education

Master of Technology (M.Tech), Embedded Systems Engineering Institute of Technology, Nirma University, Ahmadabad, India.	Jun 2018 – Jun 2020
Bachelor of Engineering (B.E), Instrumentation and Control Engineering Atmiya Institute of Technology, Gujarat Technological University (GTU), Rajkot, India.	Jun 2012 – Jun 2016